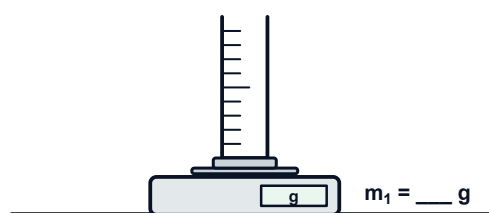


Density of a liquid

A student measures the density of a liquid.

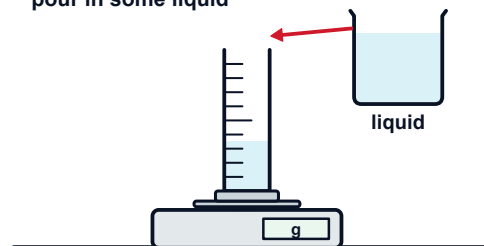
- 1** Put an empty measuring cylinder on a balance and record its mass.

empty cylinder

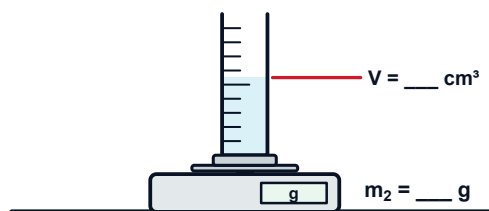


- 2** Pour some of the liquid into the measuring cylinder.

pour in some liquid



- 3** Record the volume of the liquid and the new mass on the balance.



- 4** Subtract the two masses to find the mass of the liquid.

$$m = m_2 - m_1$$

(with liquid) – (empty cylinder)

m_2 = mass with liquid, m_1 = empty

- 5** Calculate the density: mass ÷ volume.

$$\rho = m \div V$$

density (g/cm^3) = mass (g) ÷ volume (cm^3)

m = mass of liquid, V = volume of liquid